

chase 3D filaments from any available brand.

Reddy says, "Both come with their own share of advantages and disadvantages. If a buyer prefers to go with a locked printer, he or she should check if filament supply and after-sale service is available."

Filament is the input, consumable material for a 3D printer. To make a 3D model, the filament is melted and deposited in layers based on the design supplied to the printer. Filament for a 3D printer is as essential as an ink cartridge is for a 2D printer.

Reddy notes, "In locked-filament 3D printers, a chip is embedded in the filament such that the printer recognises only the filament manufactured by the same company or an authorised company."

About 80 per cent to 85 per cent of printers in the market are open printers, for which one can buy the filament from different vendors. Printers such as Ultimaker or UP BOX are open printers. Most Indian manufacturers are making open printers.

Build volume, after-sales service, output quality and speed important parameters to consider

For printers priced under ₹ 100,000, an important parameter to consider before buying is the build volume of the printer.

Next would be the number of extruders(single or dual). Reddy says, "Most printers in the Indian market come with one or two extruders only."

Accuracy, speed and resolution of the printer are also crucial. He adds, "These are highly interdependent

TABLE III
Notable Printers Manufactured in India Under ₹ 100,000

Company	Printer	Features	Price	Warranty
KCbots	Gbot 3D creator 	- Auto-levelling to automatically adjust bed height - Multi-colour printing with smart filament swapping allows stopping printing in middle and filament change - Fitted with metal hot-end nozzle made with precision machining - LCD and SD card compatibility for PC-less safe operation	₹ 66,666*	One year
LBD Makers	My3DBot 	- Has anti-choking mechanism as well as light extruder for high-speed 3D printing - Print without using PC by inserting SD card and see detailed progress on LCD - With 0.4mm brass nozzle, you can print up to 50 micron layer height	₹ 90,000	One year
Redd Robotics	FabX 	- Boasts of super precise HiWin miniature series rails - Features FDM technology, build volume of 100x100x100mm and speed between 30mm to 100mm/s, 0.4mm nozzle diameter and one extruder	₹ 19,999	One year
J Group Robotics	Dimension Delta Mini 	- Build size: Bed size 150mm, height 150mm - Upgradable to dual head - Multi-material nozzle	₹ 60,000*	One year limited warranty on mechanical parts
KCbots	Gbot Mini 	- DIY kit - Modular structure, easy to assemble and upgrade - Build volume 80x80x80mm, suitable for faster prints	₹ 25,000*	One year
Divide By Zero Technologies	AccuraCraft Studio 	- High-speed printing up to 250mm/sec - Multi-material printing: ABS, HIPS, PET, PLA - Auto-resume (resumes print after power cut)	₹ 63,000	180 days part replacement warranty

*Price excluding taxes and shipping charges

Advice to buyers

If you are a beginner and plan to use a 3D printer, go for a low-cost one initially and understand it better. Subsequently, with a strong hold in designing, invest in 3D printers with better functionality and technology.

As per our experience, look beyond common known brands. As this whole industry is new, first movers received good publicity. But there are many 3D printer manufacturers who have built amazing printers and are in the process of establishing their brands. We suggest you take expert advice before purchasing one.

—Raja Sekhar Upputuri, co-founder, Think3D

parameters because if accuracy is higher, speed can be lower."

Vishal points out that, currently, speed is a misleading parameter to look for in FDM 3D printers under ₹ 100,000, because almost all printers have similar speeds. "We have to keep in mind that 3D printer is a rapid prototyping machine and not a substitute for mass production. It will

boost conventional mass production as more ideas get physical manifestation, thereby increasing desktop manufacturing," he says. "Reliability is another important factor to consider," he adds.

Gandhi feels that quality of the printed object and corresponding speed are the most neglected factors while purchasing a 3D printer. He says, "The quality is dependent on